

Kenghou Hoi

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Research Interests

Aerial manipulation; tactile sensing; robot perception and control; motion planning; reinforcement learning; autonomous aerial systems.

Education

Zhejiang University, College of Control Science and Engineering Sep. 2024–Present
M.S. in Control Science and Engineering, advised by Prof. Fei Gao
Research focus: aerial manipulation, UAV motion planning and control, and reinforcement learning.

Beijing University of Posts and Telecommunications, School of Information and Sep. 2020–Jun. 2024
Communication Engineering
B.Eng. in Communication Engineering.

Publications

- [1] **Kenghou Hoi**, Xinqi Liu, Hongliang Huang, Junjie Wang, Anke Zhao, Jiarui Zhang, Si Liu, Juncai Long, Xin Zhou, Jialiang Hou, Yuze Wu, and Fei Gao. “Learning Robust Aerial Grasping from Monocular RGB Vision via Reinforcement Learning.” *IEEE Transactions on Robotics (T-RO)*, under review.
- [2] **Kenghou Hoi***, Yuze Wu*, Annan Ding*, Junjie Wang, Anke Zhao, Chengqian Zhang, and Fei Gao. “A Precise Real-Time Force-Aware Grasping System for Robust Aerial Manipulation.” *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.
- [3] Junjie Wang*, **Kenghou Hoi***, Yuze Wu*, Tianyu Zhao, Jialiang Hou, and Fei Gao. “A Fully Onboard Integrated Aerial Grasping System for Moving Targets in Cluttered Environments.” *IEEE/ASME Transactions on Mechatronics (T-MECH)*, under review.
- [4] Anke Zhao, Yuhang Zhong, **Kenghou Hoi**, Junyu Mou, Junjie Wang, Lijie Wang, Jialiang Hou, and Fei Gao. “Agile Fall Recovery for Quadrotors with Bidirectional Thrust via Reinforcement Learning.” *IEEE Robotics and Automation Letters (RA-L)*.
- [5] Yuhang Zhong*, Anke Zhao*, **Kenghou Hoi**, Mingli Zhu, Long Xu, Jialiang Hou, and Fei Gao. “Learning LiDAR-Based Dynamic Obstacle Avoidance via Time-to-Collision Supervision.” *IEEE Transactions on Automation Science and Engineering (T-ASE)*, under review.
- [6] Ruibin Zhang, Lun Pan, **Kenghou Hoi**, Zelong Xia, Jialiang Hou, and Fei Gao. “DPNet: Efficient Dead-End Prediction and Avoidance for Vision-Based UAV Navigation.” *IEEE Robotics and Automation Letters (RA-L)*, under review.
- [7] Jiarui Zhang, Chengyong Lei, Chengjiang Dai, **Kenghou Hoi**, Lijie Wang, Zhichao Han, and Fei Gao. “High-Speed Vision-Based Flight in Clutter with Safety-Shielded Reinforcement Learning.” *IEEE Robotics and Automation Letters (RA-L)*.

*Equal contribution.

Research Experience

Robust Aerial Grasping from Monocular RGB Vision Oct. 2025–Present
Zhejiang University

- Developed an end-to-end reinforcement-learning framework that maps monocular RGB observations and odometry directly to control commands, reducing reliance on object-specific geometric priors.
- Combined local point tracking with monocular depth estimation for persistent spatial feature extraction under severe occlusion and close-range view truncation.
- Integrated vision-language-model prompting and L_1 adaptive control to improve shape-aware interaction and sim-to-real robustness under aerodynamic disturbances and dynamic payloads.

Fully Onboard Aerial Grasping of Moving Targets in Clutter

Zhejiang University

Feb. 2025–Jan. 2026

- Built a fully autonomous, onboard aerial grasping system with an anthropomorphic gripper for moving targets in cluttered environments.
- Developed a two-stage path-search front end and a constrained trajectory optimizer that jointly considered visibility, collision avoidance, and grasp direction.
- Achieved robust onboard grasping of targets moving at up to 1.03 m/s under limited computational resources.

Real-Time Force-Aware Aerial Manipulation

Zhejiang University

Jul. 2024–Sep. 2025

- Developed a lightweight (556 g), low-cost aerial grasping system with real-time force feedback for handling fragile objects.
- Designed a Hall-effect magnetic soft tactile sensor array and a reference-sensor compensation method for geomagnetic interference, enabling accurate three-dimensional force estimation.
- Implemented a fully onboard control architecture and demonstrated precise grasping without an external motion-capture system.

Selected Honors and Awards

First Prize, RoboMaster 2023 University League (Heilongjiang), Infantry Combat	2023
Second Prize, 6th National Cyberspace Security Technology Competition	2021
Excellence in Engineering Award and Fourth Place, FIRST Robotics Competition Shenzhen Regional, Team 5831	2018
Gold Award, 2nd Macao Mechanical Engineering Challenge, VEX Robotics Competition	—
Champion, 1st Macao Mechanical Engineering Challenge, VEX Robotics Competition	—
Champion, FIRST Tech Challenge Hong Kong Qualifying Tournament	—

Leadership and Activities

- **Hongyan RoboMaster Team, BUPT:** Electrical and control team lead for two years.
- **Tianshu CTF Team, BUPT:** Competed in cybersecurity contests and organized technical seminars.
- **VEX/FRC and FTC Robotics Teams, Pui Ching Middle School, Macao:** Built competition robots and worked with rapid-prototyping tools including 3D printers and laser cutters.
- Member, Macau Cybersecurity and Capture-the-Flag Association.

Technical Skills

Programming	C, C++, Python, Rust, Verilog, MATLAB
Robotics and systems	ROS/ROS 2, Linux, embedded systems, PCB design, SolidWorks, computer vision, reinforcement learning
Development tools	Git, CMake, Vim, IDA Pro, L ^A T _E X
Languages	Chinese (native; Mandarin and Cantonese), English (IELTS 6.0)